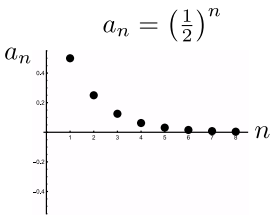
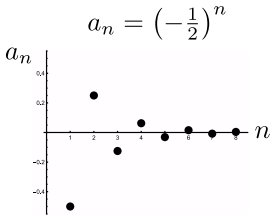
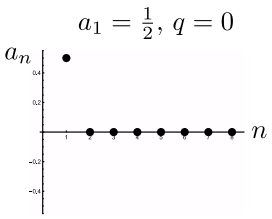
Geometrická posloupnost $(a_1 \in \mathbb{R},\ q \in \mathbb{R})$

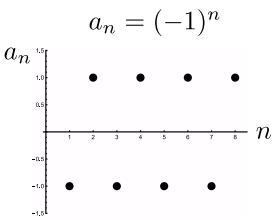
- $a_{n+1} = q \cdot a_n, \qquad n \in \mathbb{N}$ (rekurentní předpis)
- $a_n = a_1 \cdot q^{n-1}, \quad n \in \mathbb{N}$ (vztah pro n-tý člen)
- $(a_n) = (a_1, a_1q, a_1q^2, a_1q^3, \dots)$ (výčet prvků)

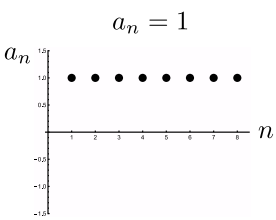
- je omezená pro $|q| < 1$ nebo $q = -1$ nebo $q = 1$



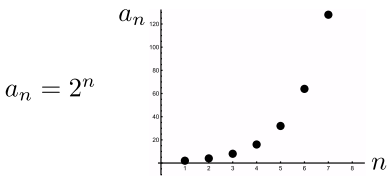


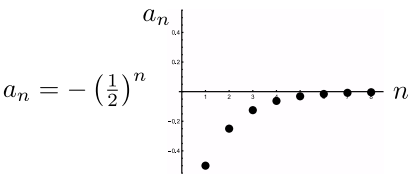




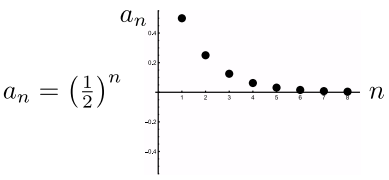


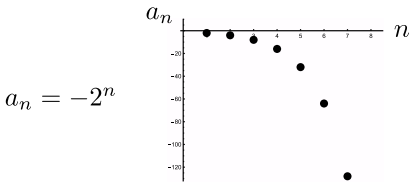
- je ostře rostoucí pro $a_1 > 0,\ q > 1$ nebo $a_1 < 0,\ 0 < q < 1$





- je ostře klesající pro $a_1 > 0,\ 0 < q < 1$ nebo $a_1 < 0,\ q > 1$





- limita
- součet prvních n-členů

$$\lim_{n\to+\infty} a_n = \begin{cases} 0 & \text{pro } |q| < 1 \\ 1 & \text{pro } q = 1 \\ +\infty & \text{pro } q > 1 \\ \nexists & \text{pro } q \le -1 \end{cases}$$

$$s_n = a_1 + a_2 + \cdots + a_n = \begin{cases} a_1 \cdot n & \text{pro } q = 1 \\ a_1 \cdot \dfrac{1-q^n}{1-q} & \text{pro } q \neq 1 \end{cases}$$

- součet všech členů pro $|q| < 1$

$$s = a_1 + a_2 + a_3 + a_4 + a_5 + \cdots = \sum_{n=1}^{+\infty} a_1 q^{n-1} = \frac{a_1}{1-q}$$

součet geometrické řady pro $|q| < 1$